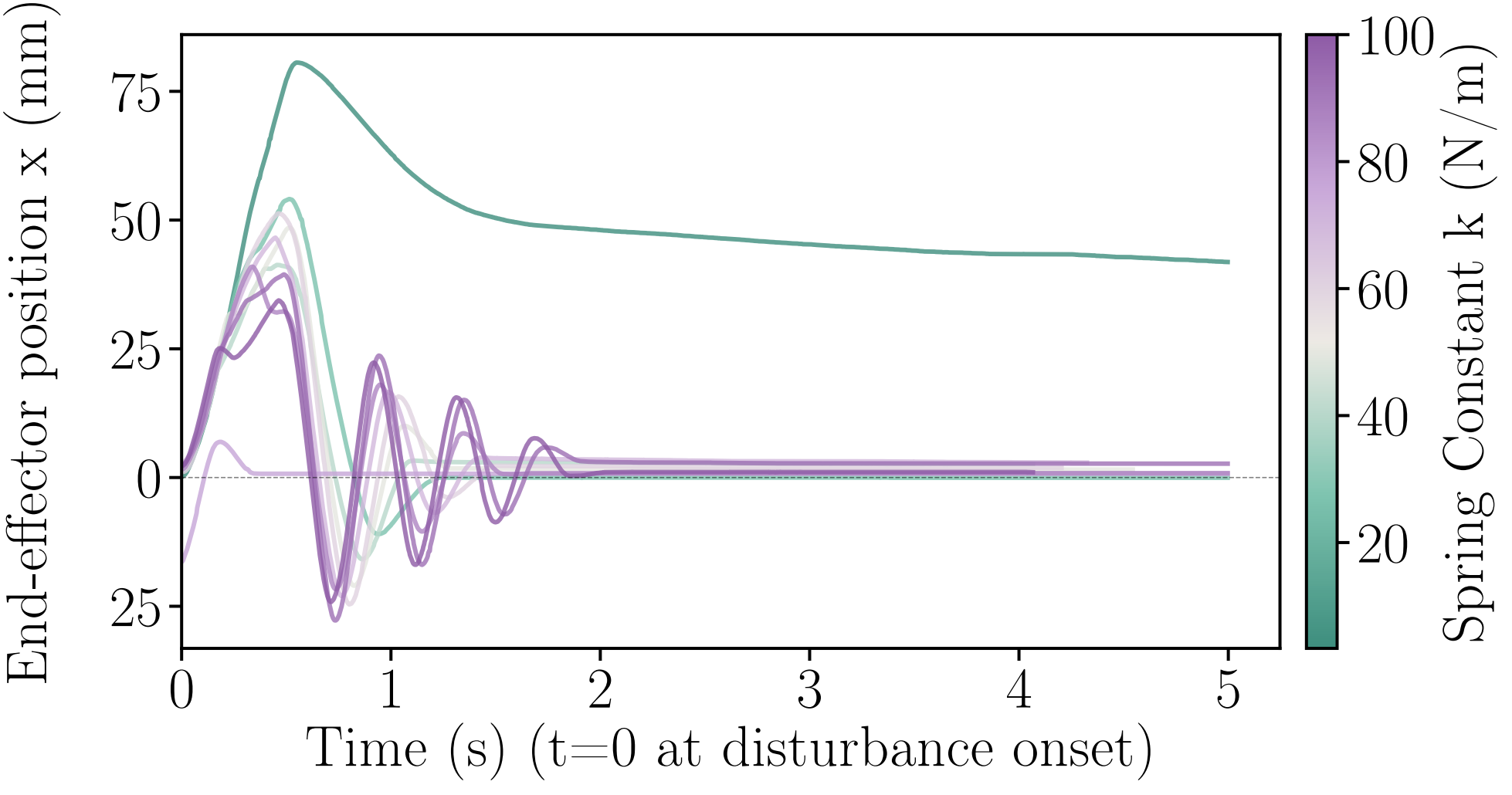

End-effector position x (mm)
75
50
25
0
25
0
1
2
3
4
5
Time (s) (t=0 at disturbance onset)
100
80
60
40
20
Spring Constant k (N/m)